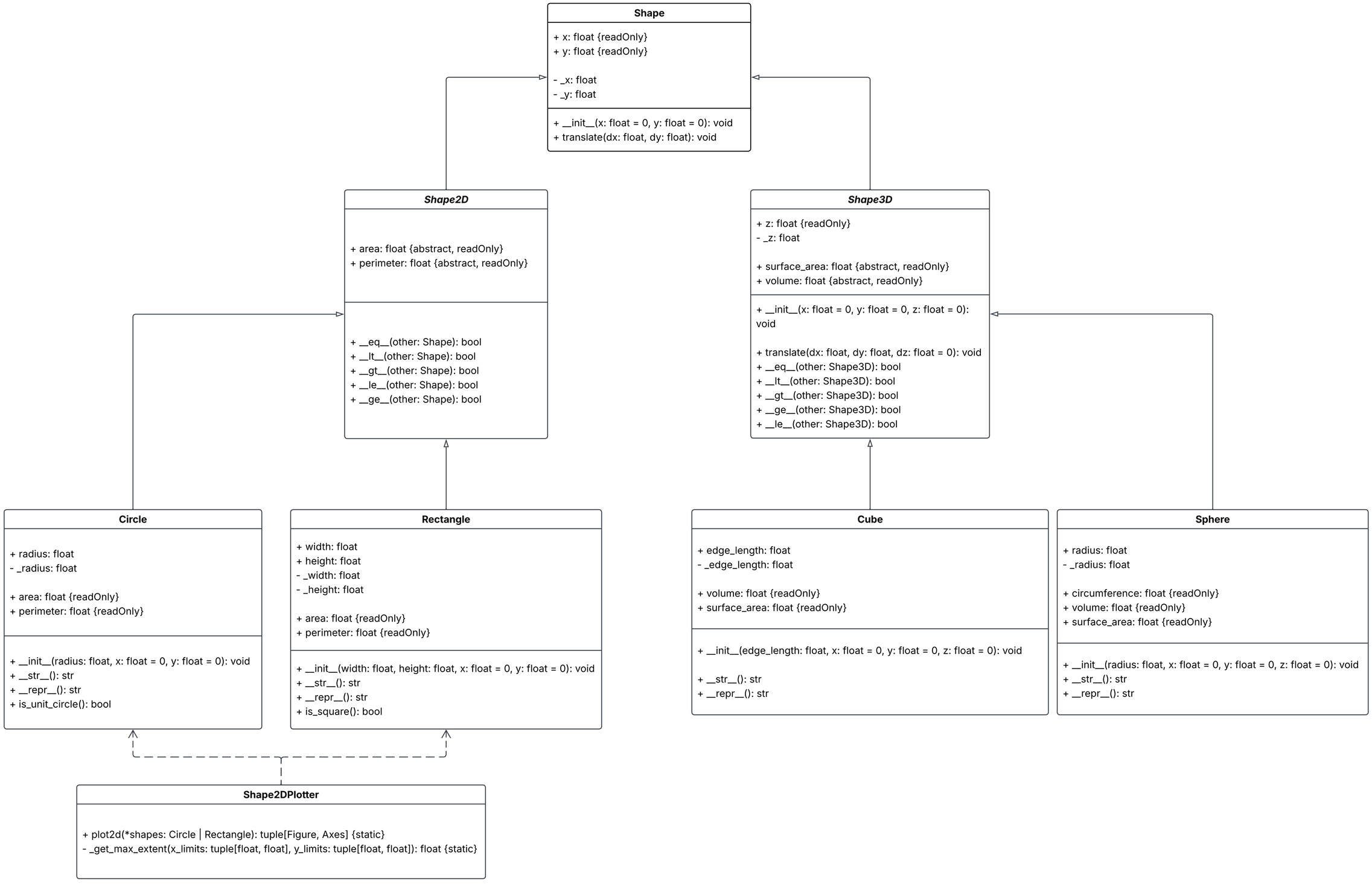

Shape
+ x: float {readOnly}
+ y: float {readOnly}
- _x: float
- _y: float
+ __init__(x: float = 0, y: float = 0): void
+ translate(dx: float, dy: float): void
Shape2D
+ area: float {abstract, readOnly}
+ perimeter: float {abstract, readOnly}
+ __eq__(other: Shape): bool
+ __lt__(other: Shape): bool
+ __gt__(other: Shape): bool
+ __le__(other: Shape): bool
+ __ge__(other: Shape): bool
Shape3D
+ z: float {readOnly}
- _z: float
+ surface_area: float {abstract, readOnly}
+ volume: float {abstract, readOnly}
+ __init__(x: float = 0, y: float = 0, z: float = 0): void
+ translate(dx: float, dy: float, dz: float = 0): void
+ __eq__(other: Shape3D): bool
+ __lt__(other: Shape3D): bool
+ __gt__(other: Shape3D): bool
+ __ge__(other: Shape3D): bool
+ __le__(other: Shape3D): bool
Circle
+ radius: float
- _radius: float
+ area: float {readOnly}
+ perimeter: float {readOnly}
+ __init__(radius: float, x: float = 0, y: float = 0): void
+ __str__(): str
+ __repr__(): str
+ is_unit_circle(): bool
Rectangle
+ width: float
+ height: float
- _width: float
- _height: float
+ area: float {readOnly}
+ perimeter: float {readOnly}
+ __init__(width: float, height: float, x: float = 0, y: float = 0): void
+ __str__(): str
+ __repr__(): str
+ is_square(): bool
Cube
+ edge_length: float
- _edge_length: float
+ volume: float {readOnly}
+ surface_area: float {readOnly}
+ __init__(edge_length: float, x: float = 0, y: float = 0, z: float = 0): void
+ __str__(): str
+ __repr__(): str
Sphere
+ radius: float
- _radius: float
+ circumference: float {readOnly}
+ volume: float {readOnly}
+ surface_area: float {readOnly}
+ __init__(radius: float, x: float = 0, y: float = 0, z: float = 0): void
+ __str__(): str
+ __repr__(): str
Shape2DPlotter
+ plot2d(*shapes: Circle | Rectangle): tuple[Figure, Axes] {static}
- _get_max_extent(x_limits: tuple[float, float], y_limits: tuple[float, float]): float {static}